Flight Discount

There are n cities, numbered 1 through n, connected by m flights. You want to travel from city 1 to city n.

You are given an array `flights` of length m, listing the flights that you can take. Each element (u, v, c) represents a flight that goes from city u to city v and has a cost of c.

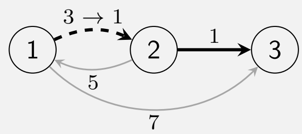
You have a coupon that can be applied to exactly one flight along your chosen route. Applying the coupon to a flight that costs c reduces its cost to $\lfloor c/2 \rfloor$.

Compute the minimum total cost to complete your journey.

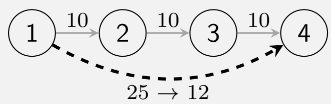
($\lfloor \cdot \rfloor$ is the *floor function*, which rounds any number down to the nearest integer.)

Example Test Cases:

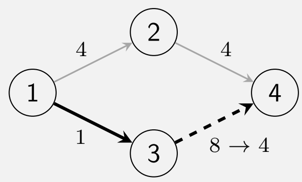
Test Case 1
Input: $n = 3$, $m = 4$, `flights` = `[(1,2,3), (2,3,1), (1,3,7), (2,1,5)]`
Output: 2
Explanation:
Flying from city 1 to city 2 using the coupon costs $\lfloor 3/2 \rfloor = 1$. Flying from city 2 to city 3 costs 1, yielding a total cost of $1 + 1 = 2$. This is the minimum cost achievable.



Test Case 2
Input: $n = 4$, $m = 4$, `flights` = `[(1,2,10), (2,3,10), (3,4,10), (1,4,25)]`
Output: 12
Explanation:
The direct flight from city 1 to city 4, with the coupon, costs $\lfloor 25/2 \rfloor = 12$. This is the minimum cost achievable.



Test Case 3
Input: $n = 4$, $m = 4$, `flights` = `[(1,2,4), (2,4,4), (1,3,1), (3,4,8)]`
Output: 5
Explanation:
Flying to city 3 and then city 4, with the coupon applied to the second flight, costs $1 + \lfloor 8/2 \rfloor = 5$. This is the minimum cost achievable.



Expected Time Complexity: $O((n+m)\log n)$

Input Format:
An integer n, an integer m, and an array `flights` of length m.

Output Format:
Return a single integer, the minimum total cost to travel from city 1 to city n.

Constraints:

1. $2 \leq n \leq 10^5$
2. $1 \leq m \leq 2 \cdot 10^5$
3. $1 \leq u, v \leq n$ and $1 \leq c \leq 10^9$ for all (u, v, c) in `flights`
4. It is possible to travel from city 1 to city n by taking one or more flights.

Instructions:

You have been provided with two template files, containing the `main` and `solve` functions respectively. The `main` function, which has been completed for you, parses the input for each test case. For each test case, it calls the `solve` function, which must compute the minimum total cost.

The `solve` function must compute this in $O((n+m)\log n)$ time.

(Refer to the `Instructions.pdf` file for detailed guidelines on the input-output format and additional test cases.)

Library Usage

For this lab, you are permitted to use any modules which are part of the standard library of either `C++` or `python`. For `C++`, do not add any extra `#include` directives; the provided `bits/stdc++.h` covers all allowed modules.